

EVALUATING REAL-WORLD PROBLEMS USING THE ORDER OF OPERATIONS – PART 1



LESSON INTRODUCTION

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

LEARNING TARGETS

- I can solve real-world problems involving the order of operations with rational numbers, including exponents and radicals.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do I solve real-world problems involving the order of operations?

LESSON NARRATIVE

This lesson builds from Lessons 7 and 8 to extend to solving real-world problems using the order of operations. Throughout the lesson, students will engage in repeated practice of using the order of operations to evaluate expressions that model real-world problems.

COMPONENT SUMMARY

12.9.1 WARM-UP (~5 MIN.)

Create an acronym

12.9.2 COLLABORATION (15 MIN.)

Apply the order of operations on expressions modeling real-world contexts

12.9.3 GUIDED PRACTICE (15-20 MIN.)

Evaluate expressions modeling real-world contexts

12.9.4 YOUR TURN! (10-15 MIN.)

Practice evaluating expressions

12.9.5 WRAP-UP (~5 MIN.)

Reflect on the muddiest part of the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Four Function Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse raising a number to the second power with multiplying the number by 2.
- Students may confuse raising a number to the third power with multiplying the number by 3.

THINGS TO CONSIDER

- Consider using data from prior lesson to determine strategic student grouping.
- 12.9.2 Collaboration students convert measurements and find volume. Encourage a discussion on the process to convert measurements first, and then finding the volume.
- 12.9.3 Guided Practice students may not recognize that they need to convert the 5 feet to inches and add to 7 to find the total height in inches. Consider reviewing unit conversion.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

12.9.1 Warm-Up provides an opportunity to use this routine. Once students have written their acronyms, collect and display them around the room. While this activity is not specific to mathematics, it is an opportunity to relationship build with students and for students to relationship build with their peers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

12.9.2 Collaboration, 12.9.3 Guided Practice, and 12.9.4 Your Turn! are grounded in students solving real-world problems involving the order of operations. Discussing real-world applications of the content will help students gain appreciation for the utility of mathematics.

DIFFERENTIATE INSTRUCTION

Different components of this lesson provide opportunities for differentiated instruction. 12.9.1 Warm-Up allows students to express their creativity in generating their own acronyms. 12.9.3 Guided Practice provides an opportunity for students to use a four-function calculator as a scaffold, so the time needed to make the calculations in the activity isn't a barrier.

12.9.1 WARM-UP: FAILURE IS AN OPTION

DIFFERENTIATE INSTRUCTION

12.9.1 Warm-Up provides an opportunity to differentiate instruction by allowing students to be creative and develop their own acronyms.

TEACHER MOVES

Consider using a Collect and Display routine once students have created their acronyms. Display the acronyms around the room and ask students if anyone would like to share theirs and explain the significance of the acronym as it relates to them.



12.9.1 Warm-Up: FAILure Is an Option

Mark Cuban is an American billionaire entrepreneur. He is the owner of an NBA team, the Dallas Mavericks, and one of the investors on the reality show *Shark Tank*.

He has been quoted as saying, "I wouldn't be where I am now if I didn't fail ... a lot. The good, the bad, it's all part of the success equation."

Failing can be when you try something that doesn't work out, but it can also be an opportunity.

1. Consider the following acronyms for F.A.I.L.

- **F**irst **A**tttempt **I**n **L**earning
- **F**orever **A**cquiring **I**mportant **L**essons
- **F**irst **A**ction **I**n **L**earning

a. Star the acronym you connect with most.

Answers will vary.

b. How can thinking about failure in this way be motivating?

Answers will vary. Thinking about failure in this way can be motivating by encouraging people to try again if they feel like they failed on the first try. Thinking about failure in this way can be motivating by providing an alternate interpretation of the word "fail," such as "the beginning of accomplishing something" instead of "the end of not doing something correctly."

c. Create your own acronym to remind you that failure is just "part of the success equation."

Answers will vary.

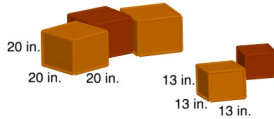
Forgive **A**nd **I**mprovise **L**ightheartedly
False **A**nswer **I** Learned



12.9.2 Collaboration: Applying the Order of Operations for Expressions in Real-World Contexts

Work with your partner to complete the following.

- Rodney needs to move some of his possessions into a storage unit. He has 3 cubic boxes that measure 20 inches (in.) on each side. He also has 2 cubic boxes that are 13 in. on each side.



To determine the amount of space all the boxes will take up, Rodney writes the following expression.

$$3 \cdot 20^3 + 2 \cdot 13^3$$

Rodney's calculations are shown.

$$3 \cdot 20^3 + 2 \cdot 13^3$$

$$\rightarrow 3 \cdot 60 + 2 \cdot 39 \quad 3 \cdot 8000 + 2 \cdot 2197$$

$$180 + 78 \quad 24000 + 4394$$

$$258 \quad 28,394$$

- Draw an arrow to the step where Rodney first made an error.
- Correct Rodney's calculations in the space next to his work.

- How many cubic inches of storage does Rodney need?
28,394 cubic inches
- Explain whether the smallest storage unit, which is 5 feet (ft) by 5 ft by 8 ft, will be large enough for his boxes.
Answers will vary. The volume of his boxes are 4.63 cubic feet and 1.27 cubic feet, which means that he will need a storage unit that can hold at least 16.43 cubic feet. The smallest storage unit, which has a volume of 200 cubic feet, is large enough for his boxes.

12.9.2 COLLABORATION: APPLYING THE ORDER OF OPERATIONS FOR EXPRESSIONS IN REAL-WORLD CONTEXTS

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 12.9.3 Guided Practice. This could include highlighting student voices to share rich discussions, or comments overhead while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.

TEACHER MOVES

Question 1b asks students to correct Rodney's calculations. Encourage students to explain Rodney's mistakes as well as their corrections.

ENRICHMENT

Students may need assistance with finding the volume of a cube. Prior to completing questions 1c and 1d, pose questions regarding the process to find the volume of cubes, such as:

- How can we find how many cubic inches Rodney needs?
- Should you convert from inches to feet and then find the volume, or should you find the volume and then convert from inches to feet?
- What other information is needed to find out if the smallest storage unit is large enough for Rodney's boxes?

12.9.3 GUIDED PRACTICE: APPLYING THE ORDER OF OPERATIONS INVOLVING EXPONENTS AND RADICALS FOR REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Consider allowing students to use a four-function calculator for this component. Students should not be able to use a scientific calculator as they would be able to enter the entire expression without considering the order. The four-function calculator can be used as an additional scaffold in this lesson so that the time needed to make the calculations isn't unrealistic.

TEACHER MOVES

Students may not recognize they need to convert the 5 feet to inches and add the 7 inches to find the total height in inches.

QUESTIONING

Question 3 involves using the Pythagorean Theorem to find the height of the television and the height that Kevin needs. Challenge the students to compare using the Pythagorean Theorem to find the missing height, or leg, to the given process to find the height that Kevin needs. Consider using visuals to assist with comparing, such as:



12.9.3 Guided Practice: Applying the Order of Operations Involving Exponents and Radicals for Real-World Contexts

- Doctors use a patient's Body Mass Index (BMI) as a diagnostic tool when analyzing their health. The following formula is used to calculate BMI, where w is a patient's weight in pounds (lb) and h is a patient's height in inches (in.).

$$\frac{w}{h^2} \cdot 703$$

Determine the BMI for an adult who is 5 feet, 7 inches tall and weighs 159 pounds. Round to the nearest tenth.
24.9

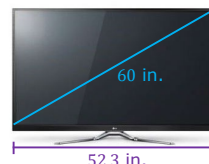
- The surface area (SA) of a cone can be found using the formula $SA = \pi \cdot r \cdot \sqrt{r^2 + h^2}$, where r is the radius of the cone and h is the height of the cone. The expression shown can be used to find the surface area of a sugar cone with a radius of 2.25 centimeters (cm) and a height of 11 cm.



$$\pi \cdot 2.25 \cdot \sqrt{2.25^2 + 11^2}$$

Find the surface area of the sugar cone in square cm. Use 3.14 to approximate π .
79.324 cm²

- Television screen dimensions are advertised using the measure of the diagonal distance across the screen. Kevin plans to buy a 60-inch television that measures 52.3 inches in width.



Kevin uses the expression shown to find the height of the space he needs to hang the television so there is at least 3 inches above and below the screen.

$$\sqrt{60^2 - 52.3^2} + 2(3)$$

Find the height of the space Kevin needs. Round to the nearest thousandth.
35.406 in.

**12.9.4 Your Turn!**

Find the value of each expression. Show your work.

$$\begin{aligned}
 1. \quad & \left(\frac{1}{5}\right)^2 \cdot (\sqrt{36} + 4) \\
 &= \left(\frac{1}{5}\right)^2 \cdot (6 + 4) \\
 &= \left(\frac{1}{5}\right)^2 \cdot (10) \\
 &= \frac{1}{25} \cdot 10 \\
 &= \frac{1}{25} \cdot \frac{10}{1} \\
 &= \frac{10}{25} \\
 &= \frac{2}{5}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & (7^2 + \sqrt{25}) - 3.75 \\
 &= (49 + \sqrt{25}) - 3.75 \\
 &= (49 + 5) - 3.75 \\
 &= 54 - 3.75 \\
 &= 50.25
 \end{aligned}$$

3. Kinetic energy is the energy of an object in motion. It can be calculated using the expression $\frac{1}{2}mv^2$, where m is the mass of an object and v is the object's velocity. If a cyclist has a mass of 71 kilograms (kg), his bicycle has a mass of 22 kg, and he is traveling at a speed of 8 meters (m) per second, the expression $\frac{1}{2}(71 + 22)(8)^2$ can be used to find the kinetic energy stored, measured in Joules.

Calculate the kinetic energy.

2,976 joules

12.9.4 YOUR TURN!**TEACHER MOVES**

If time is a concern, consider assigning 12.9.4 Your Turn! as homework.

**12.9.5 Wrap-Up: Reflection**

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on evaluating real-world problems using the order of operations?

Answers will vary. The muddiest point in today's lesson was converting inches into feet first, and then finding the volume. The muddiest point in today's lesson was knowing which values to evaluate first when finding the square root.

**12.9.5 WRAP-UP: REFLECTION****DIFFERENTIATE INSTRUCTION**

For any students that may need assistance with communicating their muddiest or most difficult point in the lesson, provide students with sentence stems, such as:

- The muddiest point in today's lesson for me was _____ because _____.
- The most difficult part of the lesson was when I _____ because I thought that _____ that _____.